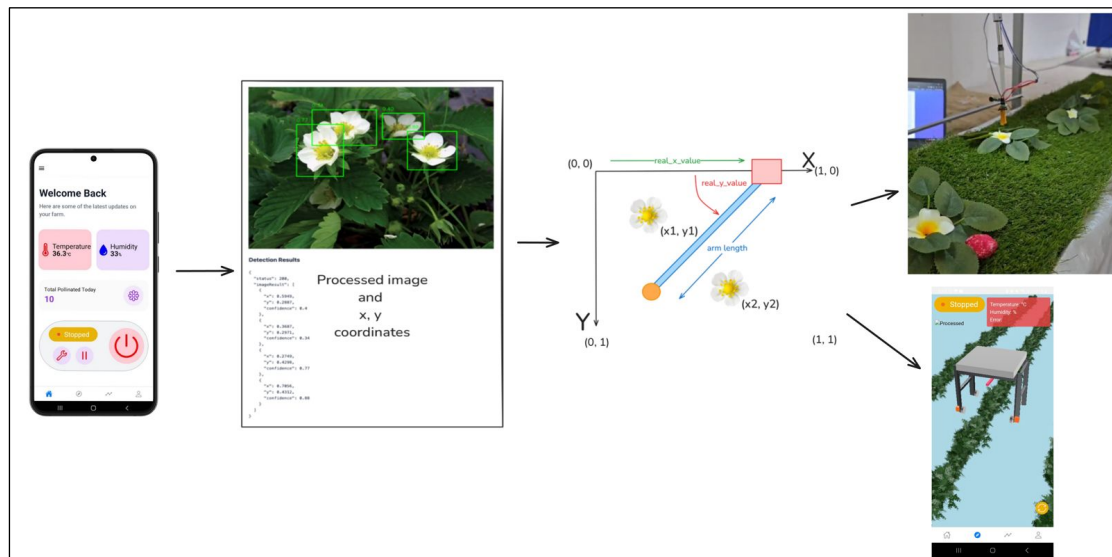


Towards Automated Strawberry Flower Pollination via IoT-Enabled Robotics

Product

This project aims to provide an alternative pollination method for strawberries by deploying a rover capable of automating the process using Internet of Things (IoT) and image processing in the farms. Strawberries are primarily grown on strips of raised soil beds. The robot is designed to travel over the soil bed, above the strawberry plants in forward and backward directions using its four legs. The central component of the rover is its robotic arm capable of three-dimensional movement. The arm is equipped with a pollinating brush at its end that is used to perform the flower pollination. The arm is designed to replicate the slight movement and vibrations caused by bees during pollination and combine it with the pollinating brush used in hand pollination to maximize the effectiveness of the process.

Methodology



Usage of Access Point connection.

ESP32 needs WIFI connection in order communicate with Main server which is hosted on locally or cloud-based solution.

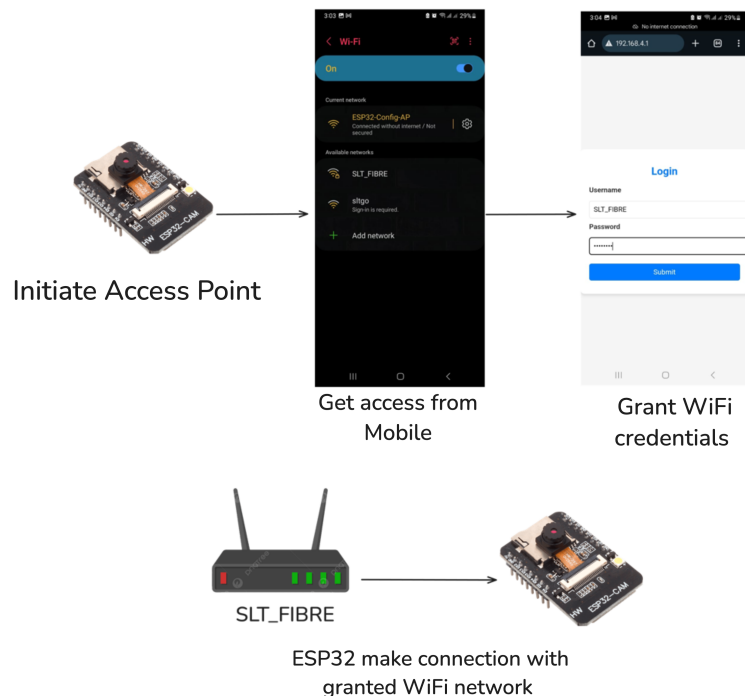
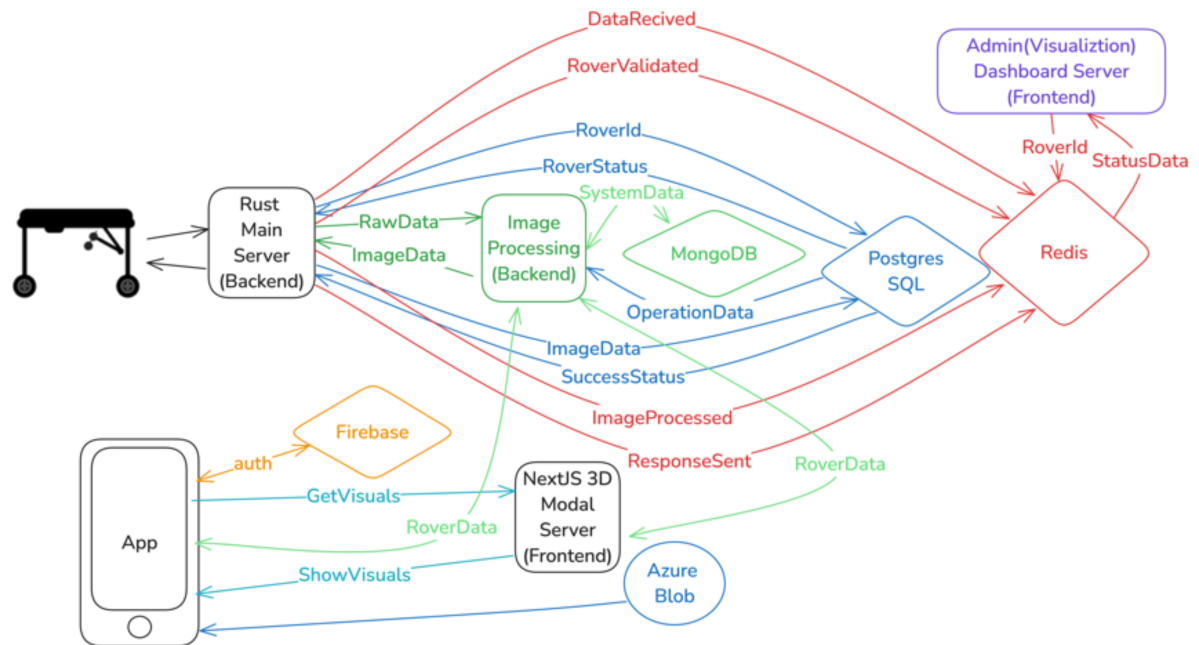


Image recognition process

- The Python server first receives a request from the rover containing the image to be analysed in base64 string format and other sensor data.
- The request is encrypted for security when receiving and must then be decrypted.
- The base64 string is then converted into an image and the base64 string and the sensor data is recorded in the database.
- The image is then given to our YOLOv8 image processing algorithm trained on strawberry flowers.
- The image processing algorithm finds the strawberry flowers and generates the x and y coordinates for the flower locations.
- The coordinates are normalized so that the coordinate system is resolution independent.
- A copy of the flower x and y coordinates are also stored in a MongoDB database along with the relevant image and sensor data. This data can then be used by the mobile app.
- Then the coordinates and any additional instructions if any are then added to a JSON file, encrypted for security and sent to the Rust server.

In the python server, all requests it receives must first go through a customized scheduling algorithm, which will then determine the execution order.

System operation with rover



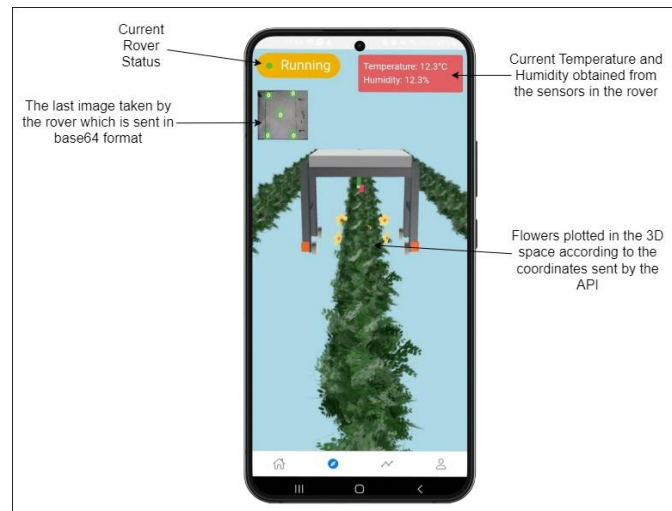
3D arm operation

First the ESP32-cam send image into Rust based server, after verifying the rover state whether it is in the operational mode, the image is sent to Python based backend server in order to recognize strawberry flowers and their x, y positions. Image recognition is done by YOLO imager processing python library and the processed x, y coordinates are sent back to Rust based server to store the data on database before send them to ESP32 as response.

Since the data from image processing modal gives x and y coordinates of the image in order to perform pollination on the 3D arm those coordinates needs to be converted by using special algorithm based on Pythagoras theory.

Usability and Accessibility

The use of a 3D simulation significantly improves the user experience, especially for farmers who may not be technologically advanced. Rather than reading through datasets or interpreting abstract metrics, they can simply observe the simulated field and rover behaviour to understand the pollination process.



Link to the Demo video

https://drive.google.com/drive/folders/1_kJiNJsvzbWzGePyaUJhy2Aw0jJPIX7u